

DengAI: predicting weekly dengue cases

Goal: estimate reported dengue cases each week in San Juan, and Iquitos using environmental measurements.

Load the data and take a first look

((1456, 24), (1456, 4), (416, 24))

1,456 historical weeks, 416 future weeks, and 24 input columns. The labels contain the city-week identifiers and the outcome.

Column	Row 1	Row 2
city	sj	sj
year	1990	1990
weekofyear	18	19
week_start_date	1990-04-30	1990-05-07
ndvi_ne	0.1226	0.1699
ndvi_nw	0.103725	0.142175
ndvi_se	0.198483	0.162357
ndvi_sw	0.177617	0.155486
precipitation_amt_mm	12.42	22.82
reanalysis_air_temp_k	297.573	298.211
reanalysis_avg_temp_k	297.743	298.443
reanalysis_dew_point_temp_k	292.414	293.951
reanalysis_max_air_temp_k	299.8	300.9
reanalysis_min_air_temp_k	295.9	296.4
reanalysis_precip_amt_kg_per_m2	32	17.94
reanalysis_relative_humidity_percent	73.3657	77.3686
reanalysis_sat_precip_amt_mm	12.42	22.82
reanalysis_specific_humidity_g_per_kg	14.0129	15.3729
reanalysis_tdtr_k	2.62857	2.37143
station_avg_temp_c	25.4429	26.7143
station_diur_temp_rng_c	6.9	6.37143
station_max_temp_c	29.4	31.7
station_min_temp_c	20	22.2
station_precip_mm	16	8.6

The environmental features come from four sources

1. Weather stations (5 features)

Ground stations measure local average, minimum and maximum temperature, daily temperature range, and rainfall. For example, **station_avg_temp_c** describes average station temperature.

2. Satellite rainfall (1 feature)

precipitation_amt_mm measures total rainfall in **mm** over a **0.25° × 0.25° grid cell**. It provides an area-based rainfall measurement alongside the station observations.

3. Climate reanalysis (10 features)

Reanalysis combines observations with a weather model to estimate environmental conditions on a consistent grid. The **reanalysis_** columns describe temperature, dew point, humidity and precipitation at **0.5° × 0.5° resolution**. Temperature levels are in **kelvin**, relative humidity in **%**, and specific humidity in **g/kg**. These describe both heat and moisture conditions.

4. Satellite vegetation (4 features)

The **Normalized Difference Vegetation Index (NDVI)** describes vegetation around each city using **0.5° × 0.5° pixels**. The columns **ndvi_ne**, **ndvi_nw**, **ndvi_se** and **ndvi_sw** refer to pixels north-east, north-west, south-east and south-west of the city centre.

Group	Column names
Location and time	city, year, weekofyear, week_start_date
Station temperature	station_avg_temp_c, station_min_temp_c, station_max_temp_c, station_diur_temp_rng_c
Station and satellite rainfall	station_precip_mm, precipitation_amt_mm
Reanalysis temperature	reanalysis_air_temp_k, reanalysis_avg_temp_k, reanalysis_min_air_temp_k, reanalysis_max_air_temp_k, reanalysis_dew_point_temp_k, reanalysis_tdtr_k
Reanalysis humidity	reanalysis_relative_humidity_percent, reanalysis_specific_humidity_g_per_kg
Reanalysis precipitation	reanalysis_precip_amt_kg_per_m2, reanalysis_sat_precip_amt_mm
Vegetation	ndvi_ne, ndvi_nw, ndvi_se, ndvi_sw

Source: [DrivenData feature definitions](#).

	city	year	weekofyear	total_cases
0	sj	1990	18	4
1	sj	1990	19	5
2	sj	1990	20	4
3	sj	1990	21	3
4	sj	1990	22	6

total_cases is the value to predict: a weekly count of reported cases. For example, the first San Juan week has 4 cases.

Check the cities and their timelines

San Juan (sj) and Iquitos (iq) have different amounts of history.

city	count	min	max
iq	520	2000-07-01	2010-06-25
sj	936	1990-04-30	2008-04-22

city	count	min	max
iq	156	2010-07-02	2013-06-25
sj	260	2008-04-29	2013-04-23

Each city's test period follows its training period: approximately five years for San Juan and three years for Iquitos. We therefore validate on later weeks.

Next, we examine differences between cities.

The cities have different case levels and seasons

city	median	max
Iquitos	5	116
San Juan	19	461

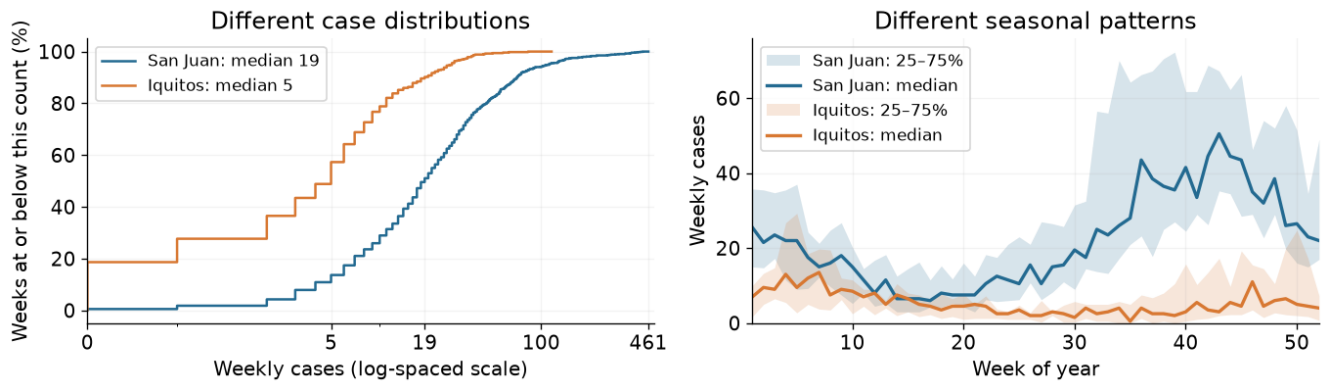


Figure 1: Case levels and seasonal timing differ between the two cities

The charts show differences in both the distribution of cases and the time of year when cases are higher.

Decision: perform EDA separately, then prepare and fit a model for each city.

Missing Data

Missing measurements across all inputs:

ndvi_ne	194
ndvi_nw	52
station_diur_temp_rng_c	43
station_avg_temp_c	43
station_precip_mm	22
dtype:	int64

Vegetation (NDVI) have the most missing values

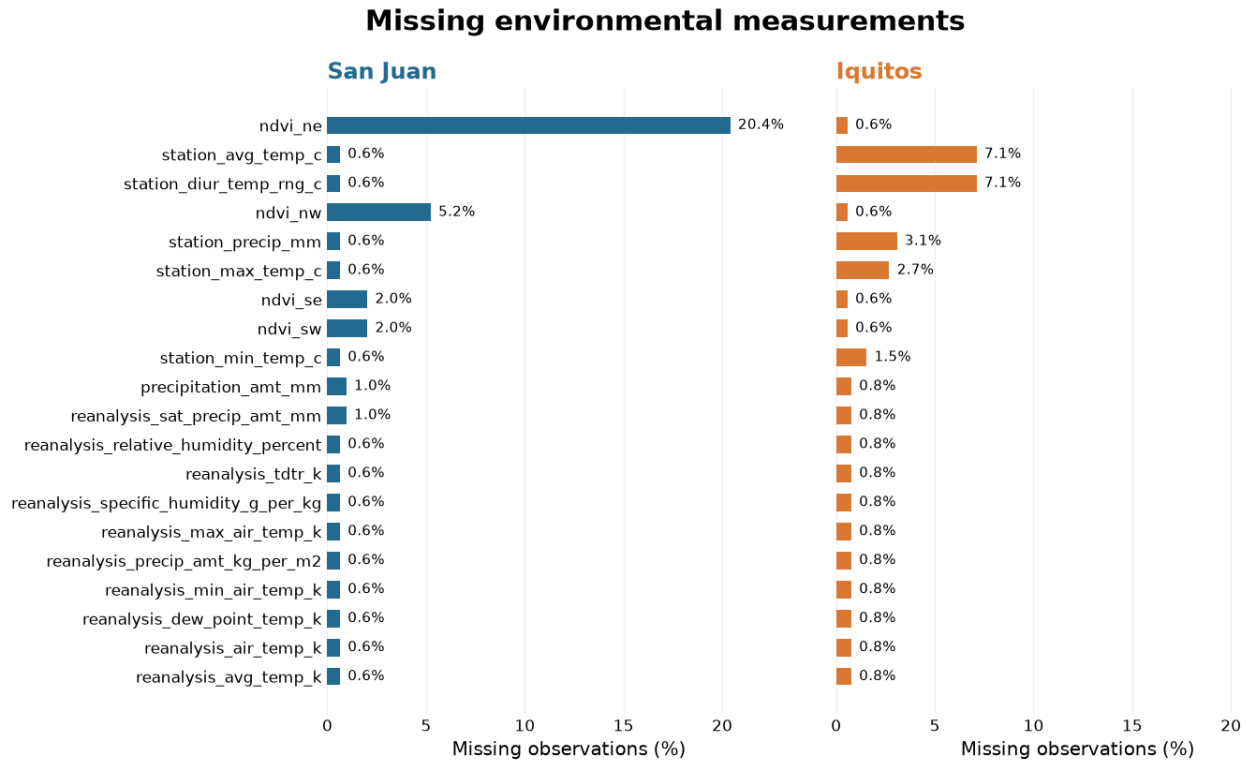


Figure 2: Missing measurements by column and city

Compare the missing percentage and longest consecutive gap within each city:

City	Vegetation feature	Missing (%)	Longest gap (observations)
San Juan	ndvi_ne	20.4	15
San Juan	ndvi_nw	5.2	15
San Juan	ndvi_se	2	14
San Juan	ndvi_sw	2	14
Iquitos	ndvi_ne	0.6	1
Iquitos	ndvi_nw	0.6	1
Iquitos	ndvi_se	0.6	1
Iquitos	ndvi_sw	0.6	1

How we fill missing measurements:

- **San Juan vegetation:** use the typical value (median) for the same week of the year, learned from the training data. If unavailable, use the overall training median for that measurement.
- **Iquitos vegetation:** estimate missing values between the nearest available measurements before and after the gap (linear interpolation).
- **Other weather measurements in both cities:** use the same interpolation approach for gaps between known values.

Why treat San Juan vegetation differently? - Its long gaps would otherwise repeat an old value for many weeks if we used forward-fill.

6. Weather history

Why look at earlier weather?

- **The effect may take time:** weather conditions today may be linked to dengue cases several weeks later. Earlier measurements help the model capture this delay.
- **Good conditions may build up:** our hypothesis is that several weeks of suitable warmth and moisture allow mosquito populations to grow more than one favourable week alone.
- **Duration matters:** rolling averages of temperature and humidity, and rainfall totals, describe whether conditions stayed favourable over several weeks.

What did we find and use?

- **In San Juan:** temperature and specific humidity have stronger positive associations with current cases around **8–10 observations earlier** than in the same week. The pattern differs in Iquitos.
- **Weather lags:** include measurements from **1, 2, 4, 8, 10, 20 and 26 observations earlier**.
- **Rolling summaries:** describe weather over the previous **4–26 observations** to capture sustained conditions.
- **Benchmark motivation:** DrivenData also suggests exploring earlier weather to address a mismatch in the timing of predicted cases. [Benchmark discussion](#)

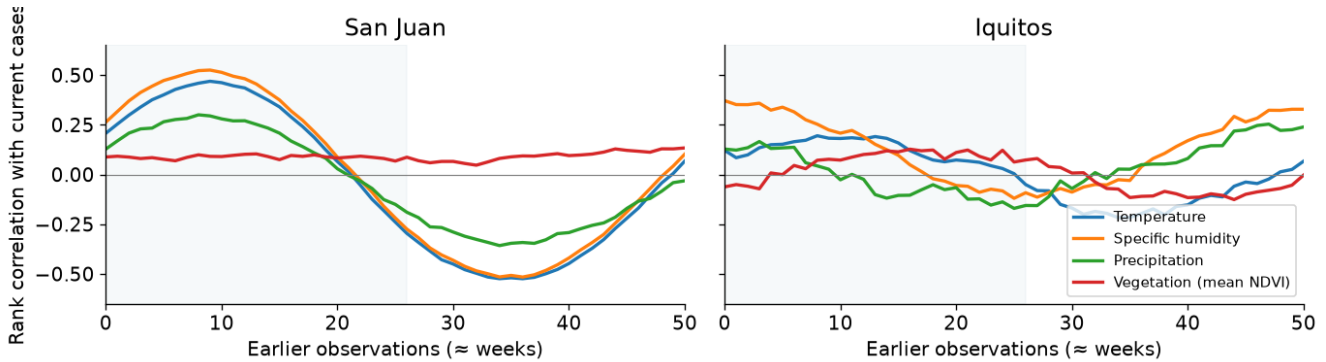


Figure 3: Evidence that weather acts on cases with a lag

7. Cross-validation

- **Keep time order:** train on earlier weeks and validate on later weeks.
- **Repeat five times per city:** predict the next **52 observations** (about one year), using more training history each round.
- **Keep a final check:** set aside the last **52 observations** as a separate holdout.
- **Choose settings using CV MAE:** lower average error is better. Then report the holdout error.
- **Why:** a random split would mix earlier and later periods. Chronological CV asks a question closer to the competition task.

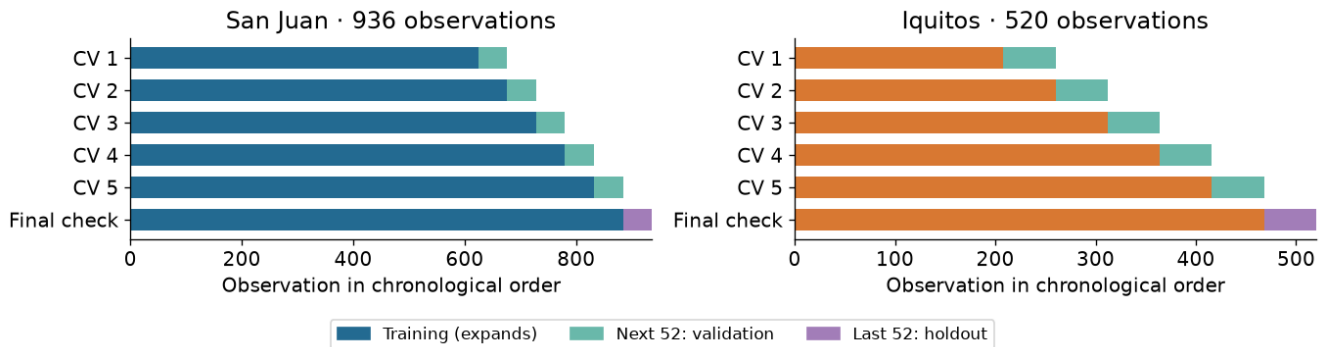


Figure 4: Chronological cross-validation timeline

8. Pipelines

- **One process:** fill gaps, add weather history, prepare inputs and train the model.

- **Avoid data leakage:** learn filling rules, scaling and feature selection from training data only.
- **Fair validation:** keep validation case counts out of training.
- **Consistent predictions:** reuse the same fitted preparation for new data.



Figure 5: The modelling pipeline